TEXORA · COURSE NOTES · LECTURE SHEET

Idea on the Global Textile and Apparel Trade

How textiles and clothing move around the world — who produces, who buys, the rules that shaped the trade, and where Bangladesh stands in it.

TE-1109 · FUNDAMENTALS OF TEXTILE ENGINEERING · TOPIC 4 · LEVEL 1 TERM 1

1. A trade as old as civilisation, an industry of global scale

Cloth has been traded over long distances for thousands of years — Indian cotton reached Rome, Chinese silk crossed Asia on the Silk Road, and Bengal's muslin was a luxury export in Mughal times. What is new is the *scale*. Today the world trade in textiles and clothing together is on the order of nine hundred billion US dollars a year (WTO figures), and clothing alone is one of the most traded manufactured goods on earth. Almost no country makes everything it wears: production and consumption are separated by oceans, and a single shirt may contain cotton from one continent, yarn spun in a second, fabric woven or knitted in a third, and sewing done in a fourth before it is sold in a fifth.

The trade has this shape because the industry is **labour-intensive at the sewing stage** and **capital-intensive at the spinning, weaving and chemical stages**. Countries with abundant, lower-cost labour specialise in garment making; countries with capital, technology and chemical industries specialise in fibres, fabrics and machinery; and the wealthy consumer markets of North America, Europe and Japan import most of the finished clothing. Understanding the trade means understanding this division of labour — and the rules and costs that hold it together.

2. The structure of the trade

Trade flows exist at every stage of the textile chain:

- **Fibres and yarns** — raw cotton, wool and man-made fibres, and yarns, move from producing countries to spinning and weaving countries. Bangladesh, for example, grows very little cotton and imports almost all of it, while India and Pakistan are major cotton and yarn exporters.
- **Fabrics** — woven and knitted fabrics are traded heavily, because many garment-exporting countries cannot yet supply all the fabric qualities their buyers demand; fabric is imported, cut and sewn, and re-exported as garments.

- **Made-ups and garments** — the finished end of the chain and the largest flow by value: garments, home textiles and made-up articles shipped to the consumer markets.
- **Machinery and chemicals** — looms, knitting machines, dyes and auxiliaries flow in the opposite direction, from the industrial countries (Germany, Italy, Japan, Switzerland, China) to the producing countries.

2.1 The leading exporters and importers

The rankings below are approximate and change slowly from year to year, but the picture they give is stable:

GROUP	LEADING COUNTRIES (APPROXIMATE ORDER)
Top clothing exporters	China (by far the largest), the European Union as a bloc, Bangladesh, Vietnam, India, Turkey, Cambodia, Pakistan, Indonesia
Top textile (yarn & fabric) exporters	China, the European Union, India, the United States, Turkey, Pakistan, Vietnam, South Korea
Top clothing importers	The United States, the European Union (Germany, France, Spain and others), Japan, the United Kingdom, Canada, South Korea

Two features stand out. First, **China dominates every stage at once** — fibres, fabrics and garments — because of its scale, infrastructure and complete supply chain. Second, several exporters, Bangladesh among them, are strong at the *garment* stage but still depend on imported fabric and yarn for part of their production; closing that gap (called backward linkage) is a central industrial question for them, and it is where much textile engineering work is found.

3. The rules that shaped the modern trade

3.1 The quota era: MFA and ATC

From 1974 to 2004, world trade in textiles and clothing was governed not by free competition but by the **Multi-Fibre Arrangement (MFA)**: importing countries set quantitative quotas limiting how much each exporting country could ship in each product category. Quotas were a restriction, but for smaller exporters they also *guaranteed market share* — a buyer who could not order more from China was obliged to look elsewhere, and this is one reason Bangladesh's garment industry grew so fast in the 1980s and 1990s. The WTO's Agreement on Textiles and Clothing (ATC) phased the quotas out over ten years, ending on 1 January 2005. Since then the trade has been formally free, and competition is decided by price, quality, lead time and compliance rather than by quota allocations.

3.2 Tariffs and preferential access

Clothing carries some of the highest import tariffs among manufactured goods in developed markets — commonly well above ten percent — so *preferential access* matters enormously:

- **Everything But Arms (EBA)** — the European Union grants least-developed countries (LDCs), including Bangladesh, duty-free and quota-free access for all products except arms. This preference is a major reason the EU is Bangladesh's largest apparel market.
- **GSP schemes** — the United States, Japan and others run Generalised System of Preferences schemes that reduce duties for developing countries, though the US scheme has historically excluded most apparel.
- **Rules of origin** — preferences come with conditions about where the fabric and yarn must come from. A garment sewn from imported fabric may or may not qualify, depending on the rule; this directly shapes sourcing decisions in factories.

3.3 Buyers, brands and compliance

The demand side of the trade is concentrated in a relatively small number of global brands and retailers — the names on high streets and online platforms — which place orders with factories in many countries and compare them continuously on **price, quality, lead time, capacity and compliance**. Compliance means the factory's conformity with labour, safety and environmental standards (building and fire safety, working hours, wages, chemical management). After the Rana Plaza collapse of 2013, in which more than eleven hundred garment workers died in Bangladesh, factory safety inspection and remediation became a permanent condition of doing business with major buyers. A factory's technical competence now includes its compliance record; the two cannot be separated.

> **KEY POINT**
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> Since 2005 there are no quotas to protect any exporter. Orders are won on the total package — price, quality, lead time, flexibility and compliance — and lost the same way. Trade literacy is therefore part of a textile engineer's professional equipment, not an optional extra.

4. Bangladesh in the global trade

Bangladesh's position is remarkable for a single statistic: it is the **world's second-largest exporter of ready-made garments**, behind only China, and garments earn more than four-fifths of the country's export income. The industry began at export scale in the late 1970s, grew under the quota system, and — against many predictions — kept growing after quotas ended in 2005. Its main strengths are:

- **A large, experienced workforce** and competitive labour cost, with particular depth in knitwear (T-shirts, polo shirts, sweaters) and basic woven garments (shirts, trousers, denim).

- **Strong backward linkage in knitwear** — most knit fabric is spun, knitted and dyed inside the country — and a growing woven and denim base.
- **Green manufacturing** — Bangladesh hosts the largest number of LEED-certified green garment factories in the world, a selling point with sustainability-conscious buyers.
- **Scale and reliability** — several thousand factories able to take very large orders and deliver them season after season.

Its main challenges are equally clear:

- **Product concentration** — a small number of basic cotton products make up most exports, while the world market is shifting toward man-made fibre (MMF) garments, in which Bangladesh is comparatively weak.
- **Imported inputs** — almost all cotton, and a significant share of woven fabric, dyes and machinery, are imported, exposing the industry to price and supply shocks.
- **Lead time** — because inputs travel far and ports are congested, total lead times are longer than in competing countries near the consumer markets.
- **LDC graduation** — Bangladesh is scheduled to graduate from least-developed-country status in 2026, after which EBA preferences will eventually be replaced by less generous arrangements; competitiveness must increasingly come from productivity and product upgrading rather than from duty savings.
- **Energy and compliance costs** — gas and electricity supply, and ever-rising buyer standards, add cost pressure that only better engineering and management can offset.

5. Where the trade is going

Several currents are reshaping the trade now. **Sustainability** has moved from marketing to regulation: the European Union is moving toward requirements on durability, recycled content and digital product information, and buyers audit chemical and carbon performance. **Man-made fibres** already account for the majority of world fibre consumption, so exporters built on cotton must adapt. **Speed and flexibility** — smaller orders, shorter lead times, rapid replenishment driven by online retail — reward factories that can plan and change over quickly. **Regionalisation** is pulling some production closer to consumer markets (nearshoring), which competes with Asian exporters on lead time rather than price. Each of these currents creates work for textile engineers: in fibre and product development, in process efficiency, in chemical management, in testing and certification, and in data-driven production planning.

6. Inside Bangladesh’s export basket

Two structural facts complete the picture of §4. First, the product split: **knitwear accounts for a little over half of Bangladesh's garment exports** — T-shirts, polos, sweaters and underwear — with woven garments (shirts, trousers, denim, jackets) making up the rest; this is unusual among major exporters and reflects the strong domestic knit backward linkage described earlier. Second, the market split: the **European Union is the largest destination** (helped by EBA duty-free access), followed by the United States and the United Kingdom, with Canada, Japan and Australia as significant smaller markets. Both splits carry the same warning: dependence — on basic products, on a few markets, and on a preference scheme tied to LDC status. Every serious policy and factory discussion in the country is, at bottom, about widening these two splits before the preferences narrow.

7. Glossary — key terms of this topic

TERM	MEANING
Backward linkage	The domestic supply chain behind an export industry — here, the local spinning, weaving, knitting and dyeing that supplies garment factories.
Quota	A quantitative limit on how much of a product may be exported to a market in a period; the MFA quota system governed textile trade from 1974 to 2004.
Tariff	A tax (duty) charged by an importing country on goods entering it, usually as a percentage of value.
Rules of origin	The conditions deciding which country a product counts as coming from, and therefore whether it receives preferential duties.
Lead time	The total time from placing an order (or receiving it) to delivering the finished goods.
Compliance	A factory's conformity with the labour, safety and environmental standards required by buyers and law.
MMF	Man-made fibres — polyester, nylon, acrylic, viscose and others — as distinct from natural fibres.
LDC	Least-developed country, a United Nations category whose members receive special trade preferences such as the EU's EBA scheme.

8. Key points and summary

- World trade in textiles and clothing is on the order of US$900 billion a year, organised as a global division of labour: fibre, yarn, fabric, garments and machinery all cross borders at different stages.

- China leads every stage; Bangladesh is the world's second-largest garment exporter, with garments earning over four-fifths of its export income, and is also the leading raw-jute exporter.
- The MFA quota system (1974–2004) shaped where the industry grew; since its abolition, orders are decided by price, quality, lead time and compliance. Tariff preferences such as the EU's EBA remain decisive for LDC exporters.
- Bangladesh's challenges — MMF products, imported inputs, lead time, LDC graduation and rising standards — are engineering and management problems as much as commercial ones.

9. Practice

Answer in your own words; brief model answers follow.

1. Explain how the MFA quota system, although a restriction, helped Bangladesh's garment industry grow — and what replaced quotas as the deciding factor after 2005.
2. A Bangladeshi factory sews garments from imported Chinese fabric for the EU market. Name the preference scheme involved and the condition (about origin) that could decide whether the garments enter duty-free.
3. Why is "backward linkage" stronger in Bangladesh's knitwear than in its woven garments, and what does that do to lead time in each case?
4. State two distinct risks created by the product and market splits of §6, and one engineering response to each.
5. A buyer moves an order from Bangladesh to a nearer, higher-cost country. Using §3.3 and §5, give two non-price reasons that could justify the move.
6. What changes for Bangladesh when LDC graduation takes full effect on EU access, and why does the sheet argue competitiveness must then come from productivity?

Answer key

1. Quotas capped the big exporters, so buyers had to place orders with smaller suppliers such as Bangladesh, guaranteeing market share while the industry learned. After 2005, price, quality, lead time and compliance decide orders.
2. The EU's Everything But Arms (EBA) scheme; the rules of origin — whether the fabric's origin lets the garment count as Bangladeshi for preference purposes.
3. Knit fabric is spun, knitted and dyed domestically in depth, while much woven fabric is imported; knit orders therefore start faster and carry shorter, more reliable lead times.
4. Product concentration in basic cotton items (response: develop MMF and higher-value products — fibre and product engineering). Market/preference dependence on the EU and EBA (response: raise productivity and quality so orders survive without duty advantage; diversify markets).

5. Shorter lead time and faster replenishment for fashion cycles; lower compliance/sustainability or logistics risk — total cost and risk, not sewing price alone.
6. EBA duty-free access is eventually replaced by less generous terms, so the duty saving that offset costs disappears; only higher productivity and better products can hold the margin.
